

Installation & Application Instructions for OSP Loose Tube Cable



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1.0 GENERAL INFORMATION

The following instructions provide general information for engineers, consultants, project managers, and construction personnel on the suitable methods for placing Loose Tube fibre optic cables. Installation equipment mentioned in this document must comply with equipment manufacturers' recommendations and regulations. Please contact AFL with questions about any method or equipment recommendations.

2.0 SAFETY

AFL strongly recommends the use of approved personal protection equipment as needed per installation. All protection equipment must meet the relevant safety requirements and all personnel involved should be aware of all applicable Safe Work Australia regulations, state and local regulations, and company policies.

2.1 WARNING

Beware of optical power exposure from light sources such as LED, VCSEL / LASER, or interfaces like optical fibre fragments. Never look directly into the end of the fibre optic cables until it is confirmed that there is no light source at the other end. There is a possibility of optical fibre fragments piercing into the skin/eyes during splicing or installation process. This might lead to infection or difficulty in removing.

Check for any toxic chemicals or fumes that can cause allergic reactions while preparing the optical fibres. Precaution needs to be taken while drilling or working closely with products containing asbestos during installation. Beware of Electric shock and Earth Potential Rise while installing near High-Voltage zones. Increased voltage in customer premises also has the capacity to damage test equipment and cablers should take reasonable precautions to prevent this from happening. Precautions must be also while installing cables with metallic elements present in it.

3.0 PRE-INSTALLATION PROCEDURE

Make sure Before You Dig Australia (BYDA) plans and the client's construction plans are on site, and all underground utilities are physically located. Identify areas that may require special consideration to install cables. Risk assessments must be carried out as a part of risk management. Machinery, equipment, and tools used for installation must be examined for proper condition of use during this time.

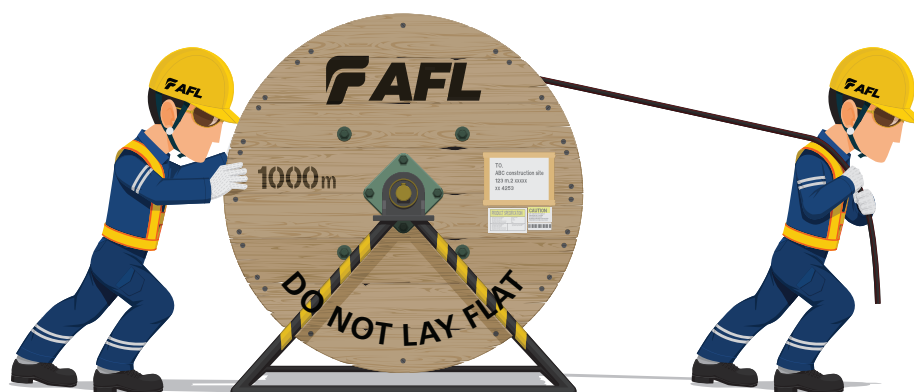
Use Job Safety Analysis or similar safety management tools to eliminate and/or control the hazards. Keep the time between the opening of the ground surfaces and the final restoration as short as practically possible. Visually inspect the cable reels for any transportation damage. Ensure that the manufacturers cable storage and handling guide is followed.



4.0 CABLE HANDLING

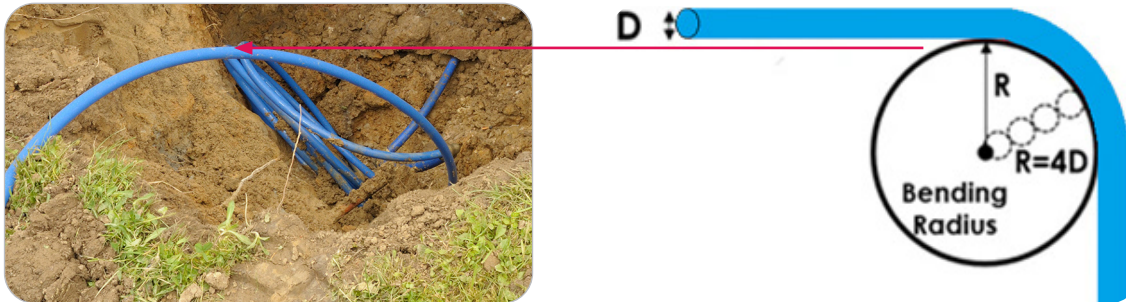
4.1 TENSION

Before the cable is pulled through for installation, it is necessary to know the tensile strength of the cable to prevent breakage of fibres and maintain long-term reliability. Exceeding the cable's maximum tensile rating can damage the cables mechanical and optical performance, void the warranty, or potentially break a fibre. When pulling or securing the cable, it is important to use the strength element(s) and not the cable jacket and to monitor the pull tension on the cable continuously using proper equipment. Always keep cable pulling tension in check and caution not to exceed the value at any time throughout the process. Please refer AFL's cable technical datasheets.



4.2 MINIMUM BEND RADIUS

Refer AFL Cable technical datasheets for minimum bending radius values during installation (Under load) and after installation (No load). These values are determined by AFL, to be the smallest bend that the cable can withstand before any increase in attenuation.



Example illustration of bending radius (R) = $4D$ of cable

4.3 KINK

Cables should never be kinked or knotted. It is important to especially monitor the cable reels during unwinding, to ensure that no mechanical damage such as kinks and/or twists occurs. Always roll the cable off the drum while de-reeling instead of spinning it off the spool end, as this will put a twist in the cable for every turn on the spool. During excess slack at pulling locations, it is recommended to lay the cable in a figure '8' to prevent cable twist. Kinking/twisting the cable damages the outer layers and places strain on the fibres, which may lead to high attenuation or even fibre breaks.



4.4 CRUSH / IMPACT

The crush performance of a cable determines the ability of the cable to not only withstand load but also recover once the load has been removed. Cables can be subjected to crush and impact under different circumstances ranging from cable ties used for installation, a hammer dropped on the cable, while hauling in duct or a vehicle driven over the cable by mistake. Care must always be taken before and during installation to not exceed the crush and impact values provided in AFL's cable technical datasheet.

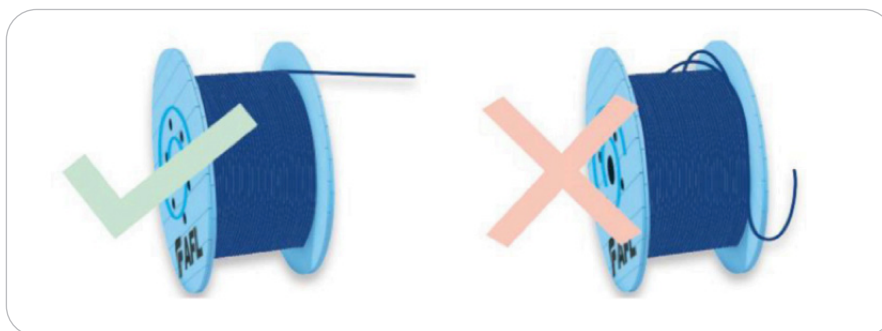


5.0 STORAGE AND HANDLING

Cable reels must always stand upright with flanges placed vertically on a flat surface. Leave enough space between stored drums for air circulation and chocks/timber blocks must be placed on either side of the flanges to prevent it from rolling. During storage, cable ends should never be left exposed.

Ensure the excess length on the inner end is safely managed. When de-reeling the cable, ensure that the inner end of the cable is released from all the attachments to the flange. Decrease the pulling tension if the cable does not payoff smoothly. For complete details and understanding refer to AFL's OSP Cable Storage and Handling guide.

All AFL cables are optically tested before they leave the manufacturing facility. Test Certificates and Certificate of Conformity by can be provided upon request. Refer AFL's "OSP Cable – Optical characteristics" for relevant fibre type of any cable design manufactured at the AFL Australia facility. Prior to installation, physical inspection and optical testing may be required based on contractors' procedure or customer requirements.



6.0 CABLE PREPARATION

Safety measures must be taken before commencing the cable / fibre stripping procedure. Ensure suitable protective personal equipment and follow local relevant safety procedures. Appropriate stripping tools must be used for easy access through the sheath layers. Try not to damage the fibres during this process.

Refer to relevant AFL's Cable Stripping guide for a comprehensive step-by-step cable stripping instruction. Fibre ends must be properly cleaved to provide satisfactory visual and geometrical end-face parameters. Use fibre end inspection tools to ensure fibres and connectors are free of debris. Ensure the tools used for cable preparation are calibrated and of appropriate size to the cable.

7.0 CONSTRUCTION LAYERS

This section details multiple elements in the cable construction that must be taken into consideration during installation.

7.1 TERMITE RESISTANT BARRIER

There are many species of termites which are economically significant in damaging buildings and infrastructure, especially in Australia, which is why a particular Polyamide (Nylon) compounds (as per AS1049) of appropriate thickness can be used against termite attack. Although termites do not derive any nutritional value from cable elements, they chew on cable when it's an obstacle in their foraging activities.

A pristine Nylon sheath will protect the cable from termite pincers as it forms a tough, shiny, and slippery barrier upon which the termite mandibles cannot get through and easily slips off. In addition to its termite resistant properties, due to its surface hardness, smooth texture, and low coefficient of friction, Nylon can be used as an outer cable sheath for cables installed through conduit with lower pulling force.



7.2 RODENT RESISTANT ARMOURING

AFL Australia employs non-metallic armouring for creating a rodent resistant barrier. Rodents attack cable for either accessibility, nutrition, or dental maintenance. Rodents attempt to feed on the cables that are in their way as their teeth grow continuously and dental maintenance is critical for their survival.

AFL employs various methods of non-metallic armouring to provide rodent resistance and robustness to the cable, whilst increasing the tensile strength. As a design principle AFL strategically place the rodent resistant layer over Nylon (termite resistant layer).

7.3 SACRIFICIAL OUTER JACKETS

The function of a sacrificial jacket is to protect the inner jacket from damage. This is quite a common practice for conventional Nylon jacketed loose tube cables which may be subjected to surface damage during the rigours of installation. If the Nylon's pristine surface is scratched, it can create a surface upon which termite mandibles can penetrate through. Polyethylene and/ or Low Smoke Zero Halogens (LSZH) are typical sacrificial sheath layers.

7.4 LOW SMOKE ZERO HALOGEN SHEATH

A Low Smoke Zero Halogen (LSZH) sheath will limit smoke emission rate when burnt and contain no toxic halogens. LSZH doesn't release dense black smoke or toxic fumes when exposed to fire. It minimises the flammability content of the cable and makes it suitable for use in indoor, free draining duct/conduit, protected (non-UV exposed) environments such as road/rail tunnels.

7.5 UV STABILISED CABLE

Exposure to sun's UV rays, will make polymers discolour over time and go brittle. It is significant to differentiate and understand whether the cables are going to be exposed to UV rays for a long-term or short-term; hence, a UV stabilised sheath must be used as an outer most cable jacket if necessary. Cable exposure to UV only during storage and transport prior to installation is considered a short-term. AFL's Application Engineers can be a resource for providing more information regarding UV stabilisation of the cable to be purchased, based on the cable application.

7.6 CABLE STRENGTH ELEMENTS

Installation techniques and varying applications place different demands on the cable. The primary design of a cable is to package the fibre in a way that minimises/eliminates the damage and stress on the fibre itself. Core to this is the use of strength elements in the cable.

A Fibre / Glass Reinforced Plastic (FRP / GRP) strength member protects the fibres from stress by absorbing the associated load. The FRP strength member is typically the central component in the cable although it can also be embedded in the cable sheath. Flexible Glass Composite strength elements, solid FRP/ GRP rods and aramids are also used in cable manufacturing at AFL Australia to provide the required strength and to retain the flexible characteristics of the cable based on their applications.

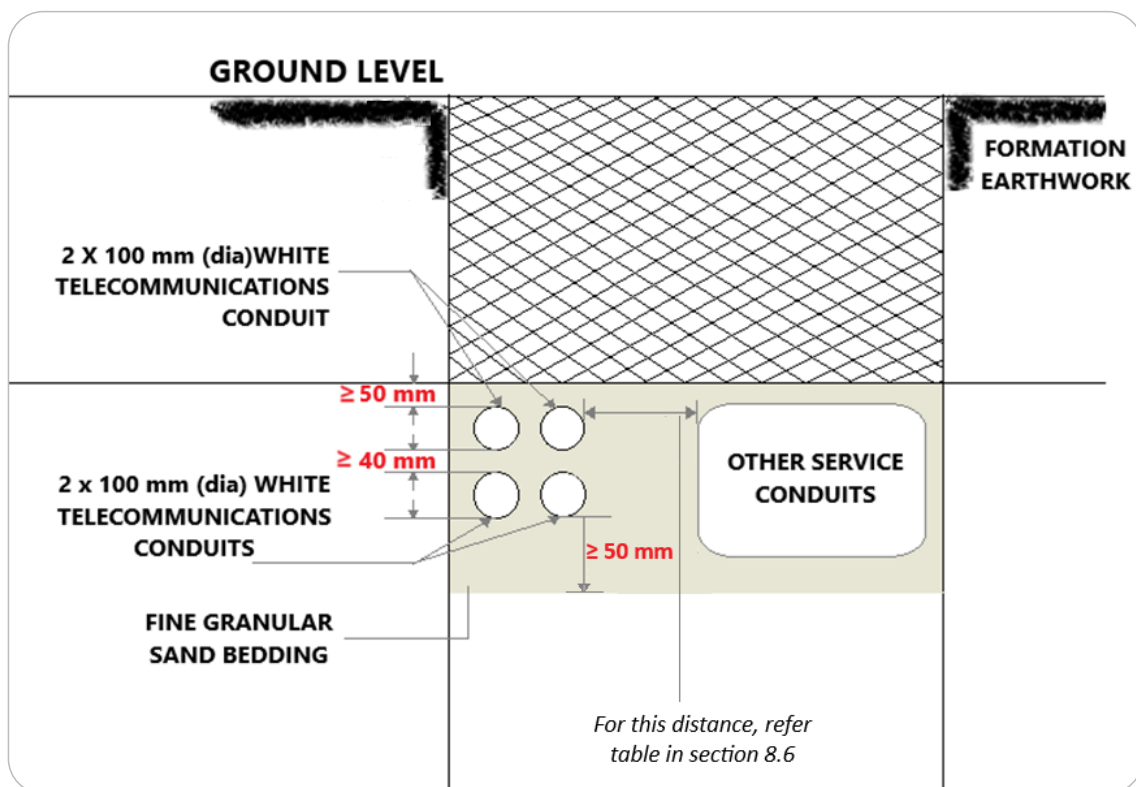
8.0 CABLE INSTALLATION TECHNIQUES

The following section provide information for installing fibre optic cables in various outdoor environments. This document also outlines installation of conduits and pits in general. Knowing the environment is the key to determine the cabling technique, thereby, ensuring a safe and efficient installation. In underground cable installations, it is important to identify the location of buried utilities and taking necessary precautions. Follow all local, state, and federal rules and regulations and safety guidelines. It is highly recommended to test for gas prior to entry and during work in all manholes and pits. If gas is detected, immediately turn of the flame if in use and get out of the pit or manhole.

8.1 CONDUITS

For new installations, grade selection, alignment (vertical / horizontal) and positioning of conduits or ducts, depends on the preferred installation technique. Try to leave the conduit ends smooth and cleanly cut without any sharp edges and burrs. Always seal any occupied conduits, where there is a history of siltation, gas, water, vermin, debris, or seal lead-in conduits, as water entry may cause flooding.

Conduits installed in clusters should be in 2 horizontal layers with lower layer having a minimum of 50mm thickness bedding using fine granular material or sand. The conduits in upper layer should be separated from the lower layer by a minimum of 40mm sand fill. Example image as below.



Conduits with Internal Diameter (ID) above 50mm are usually used for general cabling while 20mm ID conduits are used for lead-ins. Cabinets or pillars use 150mm ID conduits. According to TIA/EIA-569, the inside radius of a bend in a conduit shall always be at least 10 times the internal diameter of the conduit for all fibre optic cable installation. Ensure the cable diameter and associated bend radius are taken carefully into consideration, when designing your installation. The conduit dimensions for new or upgraded installations, bend value and their usual applications are:

CONDUIT INTERNAL DIAMETER (mm)	APPLICATION	BEND
20	Lead-in conduit	305mm radius X 90°
50	Main distribution cable routes	800mm radius X 90°
	Across road (non-distribution side of the road)	305mm radius X 90°
	Lead-in conduit	305mm radius X 90°
100	Main distribution cable routes	500mm radius X 90°
	Across road (non-distribution side of the road)	800mm radius X 90°
	Lead-in conduit	800mm radius X 90°

8.2 PITTS

Excavate a large enough hole in the required location. This is to ensure there is enough room to comfortably work in and the pit is positioned and aligned with the conduit. Appropriate amount of slack cable should be drawn into pit/ manholes for housing and jointing requirements. The length is measured from the conduit entry to the end of the housed cable. Cable length left inside pit / housing will be determined by the design parameters and the type of cable and pit/ housing used. For future cable repairs or relocation, at least 8 to 10m of slack cable must be left on average or as requested by the customer.



Spread and level a 100mm thick layer of sand or fine crushed rock at the bottom to support the pit. If power cables pass beneath the pit, install a concrete slab or approved Polymetric strip in the bottom of the hole before sand bedding for the pit. Using correct size hole saw, drill the entry holes only in the patterned area of the pit ends. Place the pit in position and ensure all the conduits entering the pit are the correct length. Level the pit with the ground surface by adjusting the sand layer. Install conduit ends inside the pit at correct depth of cover. Fit bushes to the pit or conduit end and glue to conduit.



Pits come in Plastic, Concrete, Glass reinforced concrete, and Polymer concrete types which are chosen based on the application. Number of pit-covers and number of cross bars installed between the covers varies based on the pit type selected. Dimensions of common Plastic and Concrete pit types are:

PIT TYPE	PIT NAME	DIMENSIONS (mm)		
		L	W	D
Plastic	P2	555	180	570
	P5	580	330	630
	P6	1240	440	650
	P8	1241	440	890
	P9	1920	440	890
Concrete	J6	1270	460	612
	J8	1270	460	843
	J9	1930	460	843

8.3 DIRECT BURIAL

Direct burial is a process of placing the cable underground using various techniques. Suitable cables must be selected dependent on the environmental factors such as impact, crush, bending, tensile and compressive force that may be faced during installation. AFL's Application Engineers can guide and provide technical information to choose appropriate cables based on application. Other factors that must be considered are termite and rodent resistance. Typical cable burial depth varies between 900 to 1200mm. Ensure all standards are followed for proximity to other services.

8.4 HAULING

Hauling a cable involves transferring the cable from a drum or reel into the prepared pathway. Depending on size, type and length of the cable, drum weight and dimensions may vary. Set the feeder drum at one end of the pathway and hauling devices at the other end of the pathway.

While hauling cables in conduit, a mandrel slightly larger than the cable should be hauled through the conduit first to confirm the conduit space to accommodate cable(s). While using cable guides, roller guides, conduit guides or winches, utmost care must be taken not to exceed the minimum bend radius and tensile strength of the cable as specified in AFL's cable technical datasheet. It is recommended to use water-based, non-toxic, and non-reactive lubricant, that suits the ambient temperature; in the conduit or on the cable to reduce tension exerted on the cable during hauling. To aid in fastening the cables to draw lines, a range of hauling eyes, cable grips, crimping kits and swivels can be used. Tool size must be appropriately chosen based on the type of cable under installation. When over hauling cables in existing conduits a Kevlar type or similar non-abrasive ropes should be used to not damage the existing cables. Draw appropriate amount of slack cable, usually 2 to 4m into the pit or manholes for housing and jointing requirements.

While hauling cables in trenches, cables must initially be securely attached to the pulling winch, which maintains the pulling tension. Ensure the pulling tension is within maximum pulling tension of the cable as mentioned in AFL's relevant cable technical datasheet. Rollers are placed along the trench bottom and corners (designed for bends) to facilitate the cable gliding process and to protect cable against rough excavated terrain, as shown in the below picture.



8.5 PLOUGHING

Ploughing is an efficient and non-invasive technique of cable (typically long haul).

For mass installation on a flat and open ground, ploughing is an efficient and considerably an environment friendly technique as it slits only a small section into the ground to lay the cable. It can be adopted in areas close to roads, rocks, fences, and housings.

8.5.1 DIRECT PLOUGHING

Where there are less obstacles to interrupt the cable burial process, direct ploughing can be adopted in installation areas that are open or rural. It is a speedy process that creates less disturbance to the ground compared to trenching methods. However, it requires proper direct ploughing equipment and an associated number of skilled operators. Appropriate warning tapes must be used during installation.



Mark the route with approved marker posts at all pits, road, creek, and river crossings, abrupt changes in direction and where it passes under fences. Additional markers shall be installed on curves so that there is a visual indication of the route along the curve. Maximum marker spacing is 400m in open country and 200m in rural or lightly timbered areas. Drive the marker in 600-800mm depth, nearer the property line but no closer than 600mm to the cable. Markers shall not be placed in trench or plough line, in crop or grazing paddocks, or anywhere where stock can push them over. Marker signs indicating that the cable is in the vicinity and giving a phone number for enquiries shall be attached to the marker. The recommended national range of maintenance colours are used to indicate the repeater points as mentioned in the table below. The bands are to be painted close to the top of the marker post.

ITEM OF PLANT	COLOUR OF BAND
Pit, Joint	Two Black Bands - 50mm Wide
Elevated Joint	No Marker, no colour code
ELEVATED JOINT + EQUIPMENT	COLOUR CODED MARKER AS BELOW
CPAS Contractor Alarm / Pressure Transducer point	Red Band - 100mm Wide
CPAS Test Point	Green Band - 100mm Wide
Repeater and Pair Gain Systems	Black Band - 100mm Wide
Loading Coil	White Band - 100mm Wide

8.5.2 DUCT PLOUGHING

The route must be void of any obstacles before deploying the cables. Therefore, pre-ripping must be done a minimum of 100mm deeper than the actual installation depth. Duct ploughing is done after the pre-ploughing procedure, in one go using a static or a vibrating plough. It is usually recommended to use vibrating ploughs for its easy ground penetration and refilling capacity. Backfill and compact the furrow above the installed conduit and excess soil is to be neatly heaped onto the furrow. Reasonable steps shall be taken to prevent soil erosion especially in location where both Ploughing and Trenching activities take place.

All conduit entries and exits are to be capped and left buried just below the surface and marked. Deployment of multiple ducts or conduits can be done at the same time and a trace wire is preferably placed on the top.

8.6 TRENCHING

Trenching technique is typically chosen in urban and suburban areas where number of sidewalks and street crossings makes ploughing impractical. Cables are installed in ducts of appropriate depth levels by creating a narrow passage through manual or mechanical soil excavation along sidewalks, driveways, and pavements. Hand digging of trial holes or pot holing is the reliable method to locate especially around major tree roots before continuing trench. Machine trenching can be used for any section for conduit installation only when exact location of existing electrical, gas or other services and safe clearance are established. Shoring will be used where necessary to prevent the trench from collapse and must comply with statutory regulations. Benching and battering of excavation walls can minimise the risk of soil or rock slipping onto the excavation.

In trenching method, conduits installed greater than 100mm diameter must be laid on a 50mm thickness of fine granular material or sand bedding and be surrounded by same material of similar thickness prior to backfilling.

Backfill should be compacted in layers of 150mm maximum thickness. For a wide trench, a mechanical rammer can be used. Ramming by hand tool can be done in narrow trenches. If a heavy wet clay is encountered, it is recommended to use a 50mm thick layer of sand or dry granular material on top of each successive clay layer.

In the shared trench alignment, all distribution conduits are to be installed at 450mm depth of cover. Depth of cover may also vary while continuous or non-continuous rock or obstacles are encountered underground. Where it is not possible to use shared trench, telecommunication cables must be installed in an exclusive trench. The depth of cover required under larger road or rail systems and other special areas will be determined by respective controlling authorities.

MINIMUM SEPARATION DISTANCE OF TELECOMMUNICATION CABLES (mm)			
From Low Voltage		From High Voltage	
With Protective Covering	Without Protective Covering	With Protective Covering	Without Protective Covering
100	300	300	450

Underground warning tapes / tracer wires must be used especially in shared trenches, to locate different service lines to avoid damage or interruption to any assets by future excavators. Appropriately graded, printed, and coloured warning tapes must be used depending on the service, e.g. Yellow for gas line, Green for Telephone or fibre optic cable, Blue for water pipe, etc. The recommended minimum distance that an underground warning tape be buried from the top of the ground level is 30cm. There should be a recommended minimum distance of 60cm to 90cm separation between the warning tape underground and the buried utility.



8.7 DRILLING

Drilling method is preferred while deploying cables in ducts under road or water. Possible obstacles such as existing cables, water pipes or sewers under the ground are often identified by performing a survey with ground radar in accordance with Before You Dig Australia (BYDA). Drilling is usually performed in a team of minimum two persons to locate the device and perform drilling action at the same time.

While using directional drilling equipment it is recommended to use manufacturer's approved Electrical Strike Alert System including earth mats. All staff working in the vicinity and operating the drilling machine must be wearing rubber insulated boots and gloves to the standards recommended by the manufacturer. Ensure that environmental issues that include, but not limited to, contamination by way of hydraulic oil spills, contamination of waterways, wastewater management, vegetation/tree damage and noise pollution are dealt in accordance with the relevant legislation. It begins by drilling a small diameter hole at the entry side of the project sight and eventually, acquiring the required diameter hole using appropriate reamer. Drill or bore at depths greater than 600mm in the vicinity of established trees to avoid harming tree root systems. Diameter of the bore hole should be large enough to allow for easy insertion and pulling of the specified conduit. When excavating bore entry pits through concrete or bitumen, it is recommended that only minimum breaking is undertaken. Pilot holes must be made to confirm the extent of rock.



8.8 JETTING

Cable jetting is the process of placing a fibre optic cable into an existing duct with the use of specialised installation equipment along with a high-volume air compressor. This type of installation uses a high pressure / low volume source of air to help pull the cable through the duct with the use of a Cable Carrier (A) or Parachute (B).



A



B

The installation process starts by properly attaching a fibre optic pulling grip, break-away swivel and cable carrier or parachute onto the end of the cable. The cable is then properly placed into the equipment and tractor drive mechanism. The air pressure is then applied which causes the carrier or parachute to build up pressure and form a seal. Both the speed of the drive mechanism and the amount of air pressure is then independently adjusted so that they both work together throughout the installation. The cables overall pulling force should not exceed the specified maximum tensile rating as per AFL's cable technical datasheet. The drive mechanism is designed to only assist the air flow in moving the cable forward and not to push or force the cable into the duct.

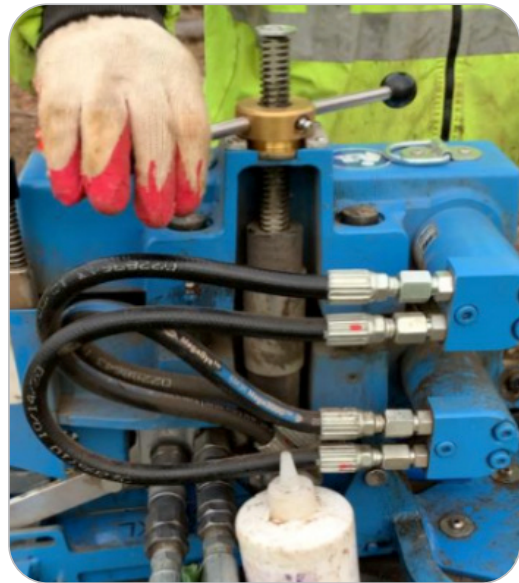
8.9 BLOWING

Blowing process use a low pressure / high-volume source of air like jetting method, but to help 'float' the cable through the duct and reduce its overall contact with the inner wall. Cable tips / end caps can also help to avoid catch points within the duct.

Determine the maximum blowing distance by analysing the route as recommended by blowing equipment suppliers. Blowing technique also referred in field as 'pushing' the cable depending on the air flow and equipment used. Piston and laminar flow are the two types of blowing unit where piston is not used in the laminar flow technique allowing complete air flow through the duct. Pushing force is higher for piston blowing unit as the resultant force has a factor of piston area multiplied to the air pressure applied.

Duct fill, cable flexibility and friction coefficient are the key factors influencing the blowing distance. It is preferred to use recommended lubricant on either the duct inner wall or the cable surface to facilitate maximum blowing distance. This process is to maintain the friction between elements for maximum throughput.

Typically, the installation process starts by properly placing the cable into the equipment and starting the tractor drive mechanism to install the cable approximately 15 to 30metres into the duct without air and under low tension. This process is used to help prevent the cable from being pushed out once the initial air is applied. The air pressure is then applied in small increments and allowed to stabilize prior to starting the drive mechanism. Once the stabilisation is complete, both the speed of the drive mechanism and the amount of air pressure need to be independently adjusted so that they both work together throughout the installation. Hand assist the payoff reel (if not powered) to reduce the tension on the installation machine and to prevent the cable from exceeding the max installation tension as specified in AFL's cable technical datasheet.



9.0 REINSTATEMENT OF SITE ON COMPLETION

Ensure that all excavation works are done to high standard with minimum disruption to the surrounding environment as much as possible. Any results that show damage or degraded characteristics of the cable under test must be rectified as per standard procedures. In any case of damage made to the property or assets in the job site, customers must be notified, and appropriate repair or alterations must be made. On completion of the work remove all plant, materials and other things that may have been brought upon the site in aid of works. Try to leave the site in a similar or better condition to how it was before the commencement of job. Ensure final sign-off on the work is received from the customer.

10.0 DOCUMENTATION

It is good practice to complete a detailed As-Built drawing and update as necessary after maintenance or repairs. Record all changes to designs and installation arrangements and forward to all stakeholders. AFL's Application Engineers can be a resource for providing information regarding previously purchased cable. Cable test reports must be part of the work completion reports. Detailed record keeping and documentation often are beneficial for future reference. Follow customer requirements and adhere to local, state, federal and industry codes, and regulations.